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Change in the active component of processed *Tetradium ruticarpum* extracts leads to improvement in efficacy and toxicity attenuation

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ABSTRACT

Ethnopharmacological relevance: The dried and nearly ripe fruits of *Tetradium ruticarpum* (A. Juss.) T.G. Hartley (**TR**) have long been used in treating headache and gastrointestinal disorders in oriental medicine. TR is usually processed by stir-frying with licorice extract before use. Although processing procedure is considered as the way to relieve pungent smell, reduce toxicity, and improve efficacy, its effects on TR's toxicity and efficacy and bioactive compound profiles are largely unknown.

Aim of the study: The purposes of the study are to evaluate the acute toxicity, efficacy and variation of toxic and effective components of TR before and after processing, and to explore the possible mechanism of how the processing procedure affect the quality of TR as a herbal medicine.

Materials and methods: Volatile oil, aqueous extract and ethanol extract of raw and processed TR were tested for their acute toxicity, analgesic, and anti-inflammatory effects in mouse models, respectively. To identify potential toxic and effective components, the extracts were analyzed with gas chromatography-mass spectrometry and ultra-performance liquid chromatography - quadrupole time-of-flight mass spectrometry, followed by fold-change-filtering analysis.

Results: LD_{50} and LD_5 tests indicated that although the aqueous extract has higher toxicity than volatile oil and ethanol extract, the use of TR is safe under the recommended does. The processing procedure could effectively decrease the toxicity of all three extracts with the largest decrease in volatile oil, which is likely due to the loss of volatile compounds during processing. Analgesic and anti-inflammatory studies suggested that volatile oil and ethanol extract of TR have better efficacy than the aqueous extract and the processing procedure significantly enhanced the efficacy of these two former extracts, whereas processing showed no substantially effects on the bioactivities of aqueous extract. Integrated analysis of animal trial and chromatographic analyses indicated that indole and quinolone type alkaloids, limonoids, amides and 18β-glycyrrhetinic acid were identified as the potential main contributors of TR's efficacy, whereas hydroxy or acetoxy limonoid derivates and coumarins could be the major causes of toxicity. Moreover, the reduced toxicity and improved efficacy of the processed TR are liked due to the licorice ingredients and altered alkaloids with better solubility.

Conclusions: In summary, the integrated toxicity and efficacy analyses of volatile, aqueous and ethanol extracts of TR indicated that the processing procedure could effectively reduce its acute toxicity in all three extracts and enhance its analgesic and anti-inflammatory effects in volatile and ethanol extracts. The promising candidate compounds related to the toxicity and efficacy of TR were also identified. The results could expand our understanding of the value of the standard processing procedure of TR, be valuable to the quality control of TR manufacturing and administration, as well as support clinical rational and safety applications of this medicinal plant.

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1. Introduction

Herbal medicines are natural-derived medicinal plants that have long existed in human history and served as an indispensable part of alternative medicine nowadays (Mahady, 2001). These plants are abundant bioactive resources of new drug/lead developments and about 40% of small-molecule agents approved by FDA are from natural origins, as proved in many cases such as camptothecin and artemisinin (Al-Hrout et al., 2017; Newman and Cragg, 2020; Pommier, 2006; Tu, 2011). Many of these natural products have pronounced bioactivities such as anticancer, antibacterial, analgesic, anti-inflammation, and antioxidant (Ashktorab et al., 2019; Hamza et al., 2018; Shan et al., 2007; Tao et al., 2020). Although the demands for herbal medicine has increased rapidly in recent years, so does the safety concerns (Ekor, 2014; Yang et al., 2017). Many basic but rather fundamental issues concerning the toxicity, efficacy, and phytochemistry of herbal medicines remain to be explored. Processing is one common practice of traditional medicine in clinics, with raw materials to be cut, cleaned, baked or stir-frying with adjuvants (such as licorice, ginger, vinegar or other herbs) according to their natural properties (Cai, 2008). This traditional practice may contribute to the change in the components of the raw plant materials, and further, their efficacy or toxicity (Gong et al., 2019; Li, Z.Y. et al., 2015b). However, the underlying processing mechanisms are yet to be investigated in detail.

The dried and nearly ripe fruit of *Tetradium ruticarpum* (A. Juss.) T. G. Hartley ("TR" for short), also named as "*Evodia rutaecarpa*" or "Wu-Zhu-Yu", is used for the treatment of headache and gastrointestinal disorders in oriental history (Editorial committee of Flora of China, 2008; Chinese Pharmacopoeia Commission, 2015). Phytochemical investigations revealed that indole and quinolone alkaloids, limonoids, and volatile oils are the main bioactive ingredients of TR. (Shoji et al., 1989; Shoji et al., 1988; Su, X.L. et al., 2018b; Sugimoto et al., 1988; Tang et al., 1996a). However, TR is labeled as "minor toxicity" in cautions recorded in some ancient Chinese medicine application guidelines. Emerging reports have shown that the overdose of TR may lead to toxic effects such as liver toxicity in rodents, hair loss, vision blur and mental disorder in some clinical cases (Cai et al., 2006; Huang et al., 2010; Li, X.Y. et al., 2015a). Therefore, for the safety and rational drug use of TR, it is important to evaluate the precise toxic and therapeutic effects of this herbal material, confirm the identities of toxic or effective ingredients, and find possible solutions to reduce toxicity while maintaining or improving its pharmacological activities.

TR is usually processed by stir-frying with licorice extract before use (Shan et al., 2020; Zhang et al., 2011). According to the description in the processing book (Cai, 2008), the processing procedure can deliver a proposed outcome of "relieving pungent smell, attenuating toxicity, and improving efficacy". Since complex physical or chemical reactions usually occur during this step, it is reasonable to assume that the alternation in herbal components is the cause of changes in herbal quality, efficacy, and toxicity (Gong et al., 2019; Zhou et al., 2017). Recent studies indicated that the volatile oil (Huang and Sun, 2012; Zhang et al., 2011), aqueous extract (Cai et al., 2014; Wang, L. et al., 2017) and ethanol extract (Li et al., 2016) of TR may all contain possible toxic compounds. Therefore, tracing the changes in chemical compositions resulted from various extraction methods combined with the processing factor may help to target the toxic or effective components.

In this study, we examined acute toxicity, analgesic, and anti-inflammatory effects of TR before and after processing with animal trial and chemical analysis. Since the volatile oil of TR (VO) is the main source of its pungent smell, aqueous extract (AE) or decoction is the common clinical dosage form, and 70% ethanol extracted compounds (EE) contains most of the major bioactive alkaloids, the plant materials were separated to the three parts accordingly, which were used in detailed evaluation. A fold change filtering analysis was also performed to identify the differentiated components that may closely associate with TR's alternation in toxicity and efficacy. The results would expand our understanding of the effects of processing on the detoxification and improvement of medical efficacy of TR, contribute to the development of proper quality control procedure in the manufacturing and administration process, and support further clinical rational and safety applications of this valuable herbal material.

2. Materials and methods

2.1. Plant material, processing product and extract preparations

The fruits of TR were collected from Yueyang County (Hunan Province, China) in August, 2018. The fruit samples were identified by Prof. Shuili Zhang, Zhejiang Chinese Medical University, and the voucher specimen (No.20180819s) was preserved at Zhejiang Chinese Medical University. *Glycyrrhiza uralensis* Fisch. (licorice) produced in Inner Mongolia (China) was provided by Zhejiang Chinese Medical University Decoction Pieces Co., Ltd (Hangzhou, China), and identified by Prof. Shuili Zhang. The voucher specimen (No.20180903) was preserved in the same place as TR. Ethanol and xylene (Huadong Chemicals, Hangzhou, China), carrageenan (Yuanye, Shanghai, China) and reference standards (Must, Chengdu, China) were used in our study; ethyl acetate, anhydrous sodium sulfate, Tween 80 (Polysorbate 80) and formic acid were provided by Sigma-Aldrich Co., Ltd. (MO, USA); MS-grade acetonitrile and methanol were purchased from Merck Chemicals Co., Ltd. (Darmstadt, Germany).

Licorice decoction was prepared by 6 g licorice pieces boiled in hot water for 30 min twice. Followed by concentration and filtering, 100 g crude TR fruits was added to the licorice extract, waiting for fully absorption, and then stir-fried until dry. The voucher specimen of the processed product (No.20180819z) was deposited at the same place as the raw plant.

The volatile oil of raw and processed TR was extracted by hydro-distillation with a Clevenger apparatus for 5 h (Hui et al., 2019). The volatile oil was dehydrated by anhydrous sodium sulfate and then filtered.

The dried fruits of TR were refluxed by a vacuum rotary evaporator (IKA, Germany) with an 8-fold volume of 70% ethanol for 1 h, and the extraction was repeated twice. The solvent was evaporated by the vacuum rotary system and the extracts were reduced to the designed concentration. For the aqueous extract preparation, the TR fruits were boiled in 10 times volume of water for 1 h, repeated with 8 times water boiled for 40 min, filtered, mixed those two extract parts and condensed to the required concentration. All the extracts were diluted by distilled water to the designed concentration before conducting pharmacological experiments.

Hereafter, we labeled the raw and processed extract samples of TR by prefix or suffix "Raw" and "Pro" as referring to extract from raw fruits of TR or its processed products. Before GC-MS detection, the VO sample was diluted with ethyl acetate (1:10) and filtered through a 0.22 μm membrane. All the AE and EE samples were diluted with methanol-distilled water (50:50) and filtered through a 0.22 μm membrane before LC-MS injection.

2.2. Animals

Male and female Institute of Cancer Research (ICR) mice 4–6 weeks (18–22 g) were purchased from the Shanghai SLAC Laboratory Animal Co., Ltd (Shanghai, China). They were acclimated 7 days before the experiment in the specific pathogen-free environment, under conditions of a 12 h light/dark cycle, temperature of 22 ± 1 °C, and humidity $50 \pm 5\%$, with standard pellet diet and water *ad libitum*. Animal care and experimental protocols were conducted according to the principles and guidelines of NIH (National Research Council, 2011), and the protocols were approved by the ethical committee of animal study in Zhejiang Chinese Medical University.

2.3. Acute toxicity test

The ICR mice were randomly divided into 1 control group and 6 treating groups (VO, AE, EE of both raw and processed TR, respectively). Each treating group contained at least 50 mice (5 males and 5 females for each dosage) according to the primary study results. VO, AE and EE were diluted by Tween 80, distilled water and distilled water, respectively, and then orally treated to mice of equal ratios dose once following 12 h fasting. The final administrated doses of each group were described in detail in Supplementary Table S2. The control group was given an equivalent volume of distilled water. The mice were observed for general behavior alternations, toxic effects, and mortality during the first 2 h, and then 4, 6, 12, 24 h after administration (Upadhyay et al., 2019). The median lethal dose (LD_{50}) and a dose at which 5% of subjects will die (LD_5) were calculated by the Bliss method (Finney, 1985). The necropsy was conducted of dead mice and some of the survivors at the end of day 14. The major organs (heart, liver, spleen, lung, kidney, and brain) were removed and fixed in 10% neutral formalin, followed by paraffin embedding and staining of sections with hematoxylin and eosin (H&E) for microscopic examination.

2.4. Hot plate test in mice

Female mice were divided into twelve groups (six mice per group). The temperature of hot plate equipment was set at 55 ± 1 °C. The reaction time of licking hind paw or jump was observed from two different directions and recorded. The cut off time is 60 s to avoid any possible injury to each mouse (Salem et al., 2019). These groups were orally treated with distilled water (Model), ibuprofen (0.18 g/kg, three times of clinical recommended dose), VO Raw and Pro (0.3 ml/kg, pre-mixed with an equal volume of Tween 80 before gavage), AE Raw and Pro (both doses of 0.75 g/kg and 1.5 g/kg, respectively), and EE Raw and Pro (both doses of 0.75 g/kg and 1.5 g/kg, respectively) after pre-treatment latency were recorded of all the tested mice.

2.5. Xylene -induced mouse ear edema

An acute inflammation model was induced on mice by xylene. Male and female mice were divided into fifteen groups (six mice each). Each was intragastrically given distilled water (Control and Model groups), VO Raw and Pro (0.3 mL/kg and 0.5 mL/kg), AE Raw and Pro (0.75 g/kg and 1.5 g/kg), EE Raw and Pro (0.75 g/kg and 1.5 g/kg), and ibuprofen (0.36 g/kg, six times of clinical recommended dose). One hour after administration, 30 μL xylene was applied to the anterior and posterior surfaces of the right ear of each mouse in all groups except for the control group. The left ear was set as control. All subjects were sacrificed after 90 min. Both ears were removed and collected the same section by an 8 mm circular punch and weighed for edema degree evaluation (Hosseinzadeh et al., 2000).

2.6. Carrageenan-induced paw edema in mice

Group and dose settings were the same as in the ear edema experiment. The right hind paw was injected 20 μL of 1% carrageenan (w/v, saline, sc) after 1 h of administration, while the control group was injected equal volume of saline (sc). The right paw thickness was measured at 0 and 4 h for all groups, and at 0, 1, 2, 3, 4, and 5 h for the EE Raw and Pro group for the potency and time course of action investigation (Solanki et al., 2015).

2.7. GC-MS analysis

GC-MS analysis was carried on a GCMS-QP2010 system (Shimadzu, Kyoto, Japan) in electron impact mode. The separation was obtained by a capillary column of DB-5 (30 m × 0.25 mm × 0.25 μm), with a programmed temperature began with 55 °C, increased to 150 °C by 2 °C/min, and increased to 250 °C at 8 °C/min, then hold for 10 min. Injection temperature, injection volume and injection mode were set at 250 °C, 1 μL and split (10:1), respectively. The ion source temperature was 250 °C and the ionization voltage was 70 eV, and the mass range was 40–550 m/z. NIST05 and NIST05s libraries were applied for components identification.

2.8. LC-MS analysis

HPLC analysis was carried on a ultra-performance liquid chromatography (UHPLC) system with a photodiode array detector (Waters, Milford, MA, USA). The separation was performed on an ACQUITY UPLC® BEH C18 column (2.1 × 100 mm, 1.7 μm) using an acetonitrile (A)-0.1% aqueous formic acid (B) system. The gradient condition was as follows: 0–0.3 min, 6% A; 0.3–3 min, 6%–30% A; 3–8 min, 30%–80% A; 8–8.5 min, 80%–95% A; 8.5–10 min, 95% A. The UV wavelength was set at 225 nm and 254 nm. The column temperature was 30 °C with a flow rate of 0.4 ml/min, while the injection volume was 0.5 μL.

The AE and EE of TR raw and processed samples were characterized by a UPLC system combined with SYNAPT G2-quadrupole time-of-flight mass spectrometry with an electrospray ionization source (Waters, Milford, MA, USA). The MS conditions were set as: capillary voltage 2.5 kV, sampling cone voltage 30 V, source temperature 100 °C with a cone gas flow of 20 L/h, desolvation gas temperature 350 °C while the gas flow of 600 L/h. Both ion modes were detected within mass range m/z 100–1200, UNIFI and MassLynx (Waters, Milford, MA, USA) were used for data acquisition and processing.

2.9. Identification and semi-quantification by high-performance liquid chromatography quadrupole time of flight mass spectrometry of TR

After acquired data of all the samples in both ion modes, raw data were transformed into mzXML files by ProteoWizard (http://proteowizard.sourceforge.net/) and followed by pair-comparison analysis in XC-MS online for fold change information (Huan et al., 2018). Then the large quantity of redundant information as well as in-source fragmentation peaks were filtered out. Subsequently, we carried out the following structure analysis by comparing it with reference chemical standards, online library, literature information, fragments, as well as software-based tools and strategy (Shan et al., 2012). Cross-validation was also conducted during the identification process.

2.10. Statistical analysis

The results were expressed as mean ± Standard Error of the Mean (SEM). The group comparisons of pharmacological effects were analyzed by ANOVA followed by Dunnett's test. The statistical analysis was performed by GraphPad PRISM 8.0 (GraphPad Software Inc., San Diego, CA, USA) and Microsoft Excel (Microsoft, Redmond, WA, USA).

3. Results

To evaluate the effects of the processing procedure on TR's toxicity and efficacy, VO, AE, and EE compounds were extracted from both raw and processed TR materials for analysis. The acute toxicity and efficacy of the six sets of samples, as well as the linkage between the potential effective components with its toxicity, analgesic and anti-inflammatory effects, were compared.

3.1. Acute toxicity of VO, AE, and EE of TR before and after processing

The same batch of TR samples was used throughout the study for consistency. The half-lethal dose range of VO, AE, and EE extract of the TR samples were firstly tested. Each group of mice was treated with equal dosage ratios of raw and processed TR for one-time oral administration (detailed doses shown in Supplementary Table S1). The toxicity responses among VO, AE and EE groups varied largely. The death time of VO groups was mainly between 6 h and 12 h, while that of AE and EE groups were within 1 h. Mice in the VO group showed acute toxic symptoms such as shortness of breath, trembling, hair colorless, muscles flabby, cyanosis, whereas in the AE and EE groups, mice had adverse effects such as diarrhea, hypokinesia and shortness of breath. In addition, mice in all the groups displayed motion disorders, loss of righting reflex and convulsion before death. LD_5 of the mice in the AE group was the lowest compared with those in the VO and EE groups, which indicated AE likely had highest toxic effects (Table 1). LD_{50} of the mice in the AE and EE groups are similar, but that in the VO group is 19.57% higher. The results also indicated that the stir-baking procedure has a great impact on TR's toxicity, with both LD_5 and LD_{50} increased dramatically in each pair of TR group after processing. Specifically, LD_5 of processed VO, AE, and EE, increased by 41.86%, 27.60%, and 9.89%, respectively. In addition, the mortality was observed increasing in a dose-dependent manner of both male and female mice (Table S1).

Histopathological analysis revealed that TR caused morphological changes in the liver, but no other main organs (Fig. 1). Severely steatosis was occurred in the liver of the VO groups, with large lipid droplets pushing the nucleus to the corner of the cell, followed by scattering hepatocytes necrotic and focal lymphocytes infiltrating. Mice in the EE group has a similar pattern of liver injury but with a lower degree. In addition, mice in the AE group showed the symptoms of cellular edema and hyaline changes, with much less scattering inflammation and steatosis comparing with those in other groups.

3.2. Analgesic effects of VO, AE, and EE of raw and processed TR on mice in hot plate test

The effects of VO, AE, and EE with both raw and processed products of TR on a classic analgesic mouse model were compared. All the treated groups prolonged the latency of responding time on the hot plate (Fig. 2A). When compared with the model group, the VO group demonstrated a significant analgesic effect at all three-time points. In the meantime, we observed that TR's analgesic effect was dose-dependent, and the EE group (1.5 g/kg, two times dose of clinical usage) had more potency than the AE and even ibuprofen (0.18 g/kg, three-fold of recommended dose clinically) groups. Moreover, the processing procedure exemplified analgesic activities in all groups and less response time was needed for TR's extract groups when compared with ibuprofen. Although the three times dose of the positive drug (ibuprofen) had no significant difference compared with the model group, it could extend latency time of mice on the hot plate. In addition, AE (1.5 g/kg, i.e. two times of clinical dose) showed similar analgesic effects as ibuprofen (0.18 g/kg). In summary, within 90 min, VO and EE have stronger analgesic activities than AE in a dose-dependent manner, and the processing factor significantly enhanced the pain-relieving effect of TR.

Table 1
LD_5, LD_{50} of raw and processed *Tetradium ruticarpum* oral administrated once in mice (g/kg).

Group	Subgroup	LD_5	LD_{50}	LD_{50} 95% confidence intervals
Volatile oil	Raw	159.8	264.1	230.3–297.4
	Processed	226.7	375.6	333.0–421.3
Aqueous extract	Raw	100.0	222.0	185.9–286.6
	Processed	127.6	230.4	206.0–305.5
Ethanol extract	Raw	145.6	220.8	194.9–251.2
	Processed	160.0	230.4	205.7–259.3

The TR's volatile oil yield of raw product is 1.17% (ml/g, n = 10), yield of the processed product is 0.86% (ml/g, n = 10).

3.3. Anti-inflammatory effects of raw and processed TR on xylene -induced ear edema in mice

The xylene-induced inflammatory model was applied to evaluate the anti-inflammatory effects of all tested groups (Fig. 2B). Results indicated that the EE, AE and VO groups all indicated the potency of decreasing ear edema. Processed EE groups at both one and two folds of clinical dose had the strongest effects of relieving edema in ears, while the raw EE group at high dose possessed similar significant activities. The pharmacological effect was also dose dependent. The results also indicated that the VO and EE groups of the processed TR treatment have better ear edema remission than those of the raw TR, respectively.

3.4. Anti-inflammatory effects of raw and processed TR on carrageenan-induced paw edema in mice

To further explored the anti-inflammatory effects of TR, carrageenan was used as a different stimulant. As the EE group (1.5 g/kg) displayed the most potency of anti-inflammation on the previous ear edema model, the EE group was chosen to conduct a time-course study on the paw edema model first. The paw edema increased gradually from 1 h after inducing, and then reached a peak degree at 4 h (Fig. 2C). The administration of EE and ibuprofen both reduced the edema level at the beginning, with the full potency of relieving paw edema symptoms at 4–5 h of EE treatment and demonstrating better anti-inflammatory effects after 4 h's carrageenan stimulating. The processed group gave a sharp fall of paw edema degree after inducing for 3 h compared with raw and positive control ones. Thus, we further performed a 4 h-paw-edema-degree observation for all the test groups. The VO and EE groups led to significant paw edema attenuating effect (Fig. 2D). On the contrary, AE indicates no anti-inflammation on the paw edema model. These results suggest that EE and VO have a dose-dependent strong anti-inflammatory effect on carrageenan-induced paw edema mouse model, while AE shows little therapeutic effect. Moreover, the processing procedure could make TR to have better anti-inflammatory potency and prolonged effect in the later-stage time course.

3.5. Analysis of volatile oil of raw and processed TR

Since the VO group has relatively lower toxicity with significant therapeutic effects compared with other groups under the same scale of converted dose and augmented effects as alleviated toxicity with enhanced efficacy after processing, the modification of volatile components after the processing procedure may play an important role in TR's efficacy and toxicity. Therefore, VO of TR samples before and after processing were subsequently analyzed.

The VO yield of raw and processed TR was 1.17% and 0.86% (ml/g), respectively (Fig. 3A), indicating that the processing procedure decreased the VO content of TR significantly. Eighty-three components were identified in volatile oil from raw and processed TR fruits by GC-MS. The representative chromatography was shown in Fig. 3B and the relative area percentages of the compounds were summarized in Table S2. The total ion chromatographs of raw and processed VO were similar. Interestingly, terpenoids and their derivatives are the typical VO constituents of TR, with β-cis-ocimene, β-myrcene, β-phellandrene, β-trans-ocimene and β-linalool counting for over 80% of the total VO content. Moreover, the proportions of β-myrcene, β-trans-ocimene and β-linalool decreased, whereas β-phellandrene increased after processing. The substantially decreased total content and component profile may related to the alleviated pungent smell, reduced toxicity, and improved efficacy of TR after processing.

3.6. Further analysis of the separated compounds to identity potential toxic and therapeutic contributors

Since the acute toxicity and efficacy varied from the AE and EE

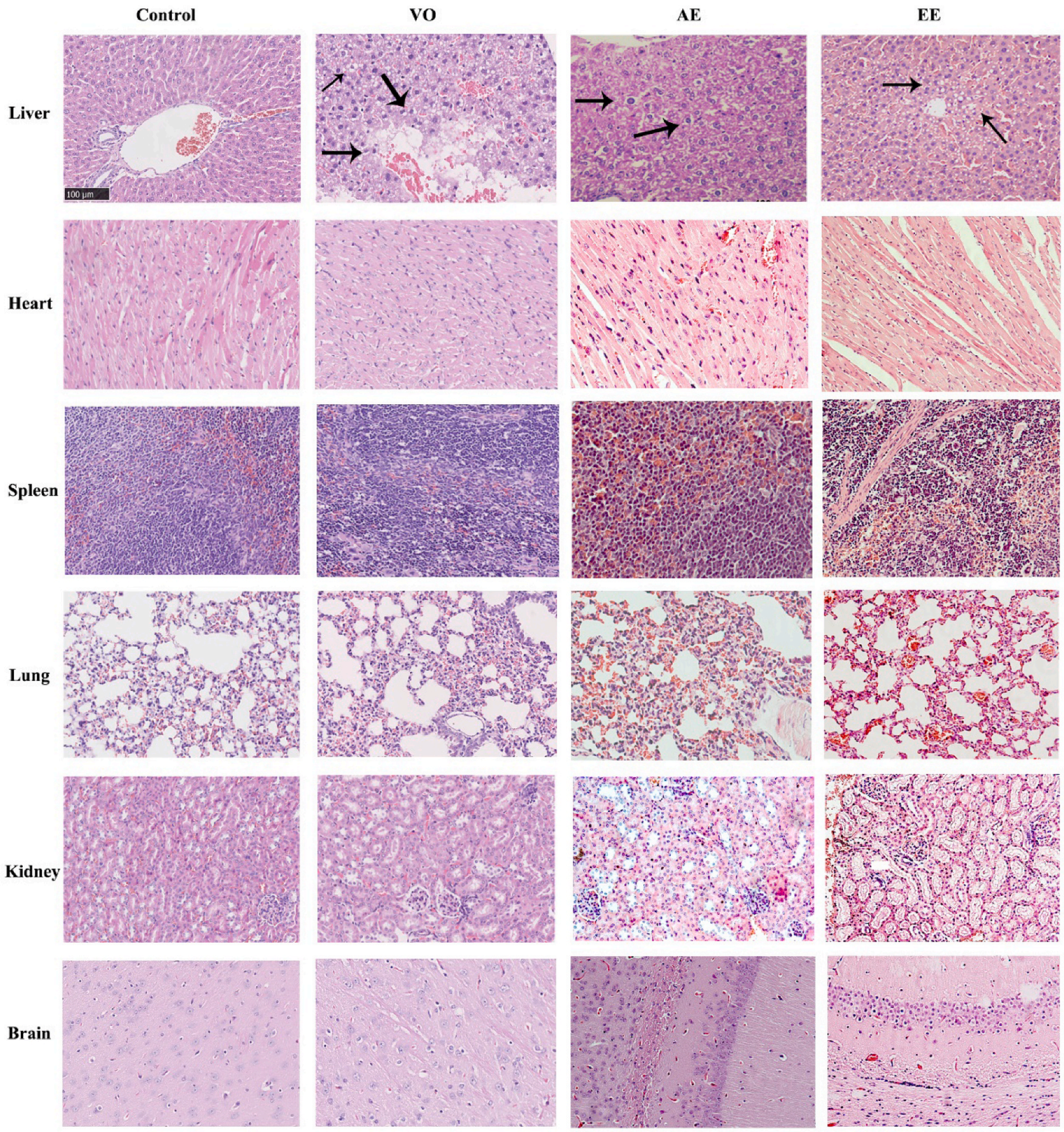

Fig. 1. Histopathological graphs of liver and other main organs in mice after one-time oral administration of control, volatile oil (VO), aqueous extract (AE) and ethanol extract (EE) of *Tetradium ruticarpum* (× 200).

groups, tracing the differentiated components from AE and EE of TR may delineate the processing mechanism. Based on the changing toxicity and efficacy activities under various extract or processing conditions, the fold change filtering strategy was applied for screening differentiated components with large fold changes during those treatments (Wang, Z. R. et al., 2017; Xu et al., 2018), followed by the identification procedure to acquire potential toxic and therapeutic contributors.

The typical base peak intensity chromatographs of reference standards, AE and EE groups of both raw and processed samples are presented in Fig. 3C. Comparisons were conducted on extract methods and processing factors. For instance, the potential toxic ingredient pool was obtained by fold change filtering of raw AE *vs* raw EE of TR (Table 2), and the potential effective component pool was acquired by fold change filtering of pro EE *vs* pro AE (TOP 60 is presented in Table 3). 6-Acetoxy-5-epilimonin, 12α-hydroxylimonin, 4-methoxycoumarin and quinic acid were the four possible toxic components based on fold change filtering and acute toxicity performance of raw TR. Interestingly, as shown in Table 3, most of the identified components among the TOP 60 of pro AE and pro EE fold changes were alkaloids, along with scattered limonoids, amides and 18β-glycyrrhetinic acid. Moreover, 70% ethanol extraction could increase the solubility of major components than water extraction, except for 6-acetoxy-5-epilimonin. The intensity of this compound decreased, which was in accordance with the potential toxic ingredient screening results mentioned above. The raw and processed factor, which also exhibited an impact on TR's components, gave a completely different result based on the fold change strategy (Table 4). Fold change comparison of raw and processed TR was conducted within AE and EE group, respectively. The ingredients from licorice attributed one-third of the changes in TOP 20 in AE and EE group, respectively. Hydroxy or acetoxy limonoid derivates, as well as coumarin content, decreased after processing within AE and EE groups, while phenolic acid and flavonoids were not significantly affected by processing.

4. Discussion

Herbal products have provided numerous successful approved drugs or potential leads towards treatment of a variety of human diseases (Hamza et al., 2020; Kamal et al., 2018; Newman and Cragg, 2020). For example, some ethnopharmacological herbs, their extracts and bioactive compounds, as well as other herbal and natural product-derived

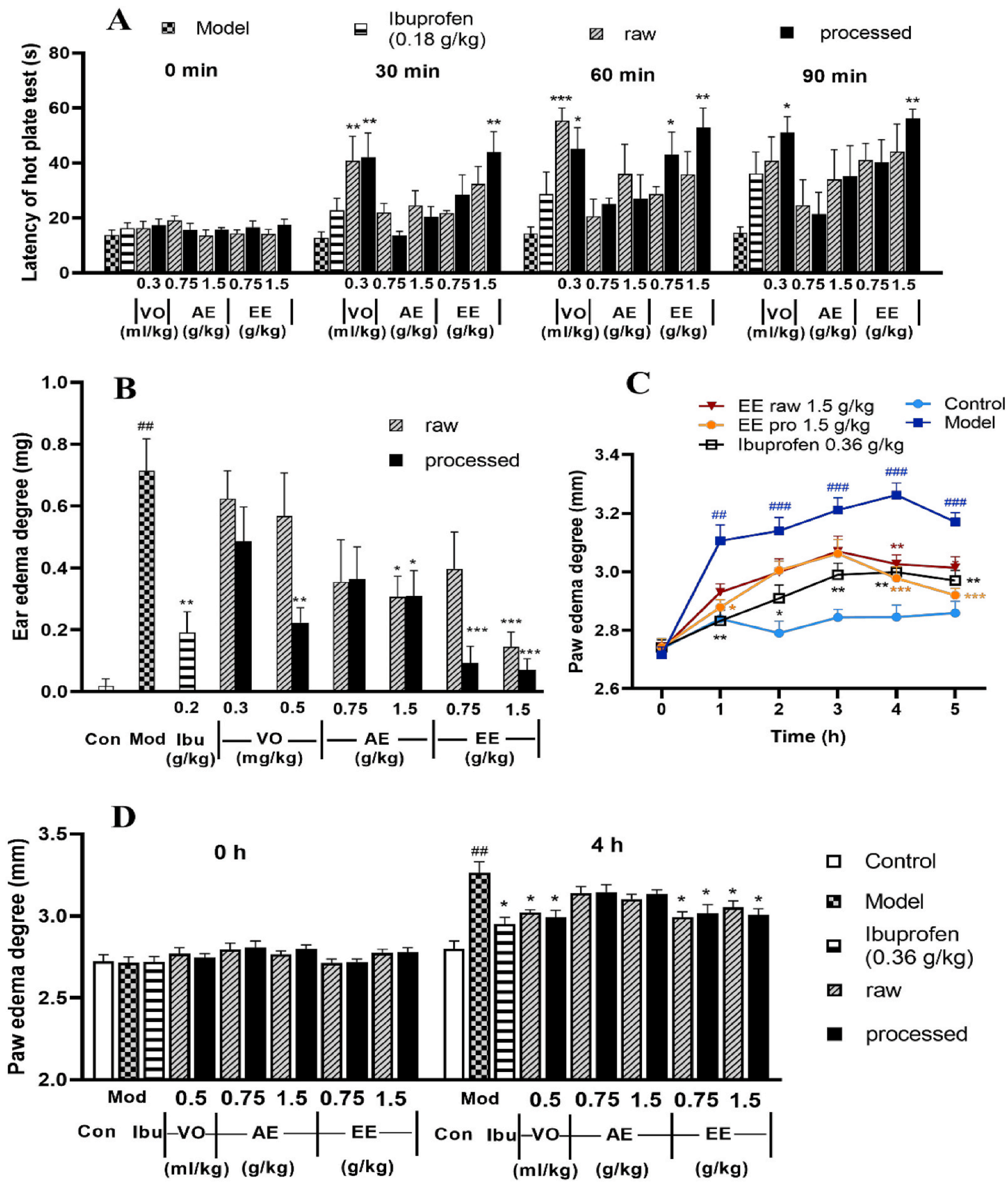

Fig. 2. Effects of VO, AE, and EE with raw and processed *Tetradium ruticarpum*. Analgesic effects of volatile oil (VO: 0.3 ml/kg, 0.5 mL/kg), aqueous extract (AE: 0.75 g/kg, 1.5 g/kg) and ethanol extract (EE: 0.75 g/kg, 1.5 g/kg) of raw and processed TR on mice in hot plate test, with Ibuprofen (0.18 g/kg, twice as clinical usage) as the positive drug, n = 6 (A). Anti - inflammatory effects of TR's VO (0.3 mL/kg, 0.5 ml/kg), AE (0.75 g/kg, 1.5 g/kg) and EE (0.75 g/kg, 1.5 g/kg) on the xylene-induced ear edema in mice after one - time oral administration, with Ibuprofen (0.18 g/kg, three-fold dose as clinical usage) as the positive drug, n = 6 (B). Time course of anti-inflammatory effects of TR's EE (1.5 g/kg) and Ibuprofen (0.36 g/kg, six-time dose as clinical usage) on the carrageenan-induced paw edema in mice for one-time treatment, n = 10 (C). Anti-inflammatory effects of TR's VO (0.5 mL/kg), AE (0.75 g/kg, 1.5 g/kg) and EE (0.75 g/kg, 1.5 g/kg) on the carrageenan-induced paw edema in mice after gavage treatment, once, n = 6 (D). ##$p < 0.01$, model group compared with control group; *$P < 0.05$, **$p < 0.01$, ***$p < 0.001$, compared with model group.

medicines, have shown promising therapeutic effects on cancer treatment and other diseases in both *in vitro* and *in vivo* study (Al-Akhras et al., 2012; Al-Dabbagh et al., 2018, 2019; Al-Hrout et al., 2018; Amin, 2008; El-Dakhly et al., 2020; Hamza et al., 2016; Wilhelm, 2006). Therefore, comprehensive potency and safety investigations on these herbal extracts and their bioactive components are crucial for their application in the future.

Although TR has drawn a great attention of researchers on its bioactivities and phytochemistry parts (Iwata et al., 2005; Ko et al., 2007; Shoji et al., 1988), the efficiency of the licorice processing procedure on its toxic and therapeutic effects, as well as the change of bioactive compounds during processing, are yet to be explored. The present study accurately evaluated the acute toxicity and efficacy of raw and processed TR extracts (VO, AE, and EE) *in vivo*, as well as comprehensively analyzed the components linking to its toxicity and pharmacological potency.

The acute toxicity of VO, AE, and EE of raw and processed TR was conducted according to the standard guideline of modern toxicity analysis. Among the VO, AE, and EE groups, mice in the AE group indicated a relatively high risk of toxicity, whereas the processing procedure could reduce toxicity in all three extracts (Table 1). Interestingly, the VO group showed substantially decrease in LD_{50} after processing, which may due to the fact of decreased VO content yield combined with its reduced toxicity (Fig. 2A and Table 1). Heating and stir-frying with licorice extract could accelerate the volatilization of volatile materials, and thus attenuate its pungent smell and reduce the toxicity remarkably. Moreover, given by the converted coefficient mouse/human as 12.3 (Nair and Jacob, 2016), TR's acute toxicity in mice (LD_5 and LD_{50} as 100 g/kg and 220 g/kg, respectively, Table 1) indicated that the estimated toxic dose for humans would be 8.13 g/kg and 17.89 g/kg, respectively. Compared with the 2–5 g/day as the recommended dose of TR, this does is at a relatively harmless level (Hayes and Loomis, 1996). Furthermore, our study revealed that the LD_5 of all groups is 100-fold of the effective dose, indicating a relatively safe therapeutic range for clinical use. Taken together, the acute toxicity of TR remains to be non-toxic in our study at the currently suggested doses, and processing could significantly further reduce the toxicity of TR in each extract group. In addition, carefully monitoring would still be needed during high-dose or long-time usage in clinics. It would be interesting to further investigate its chronic low-dose toxicity,

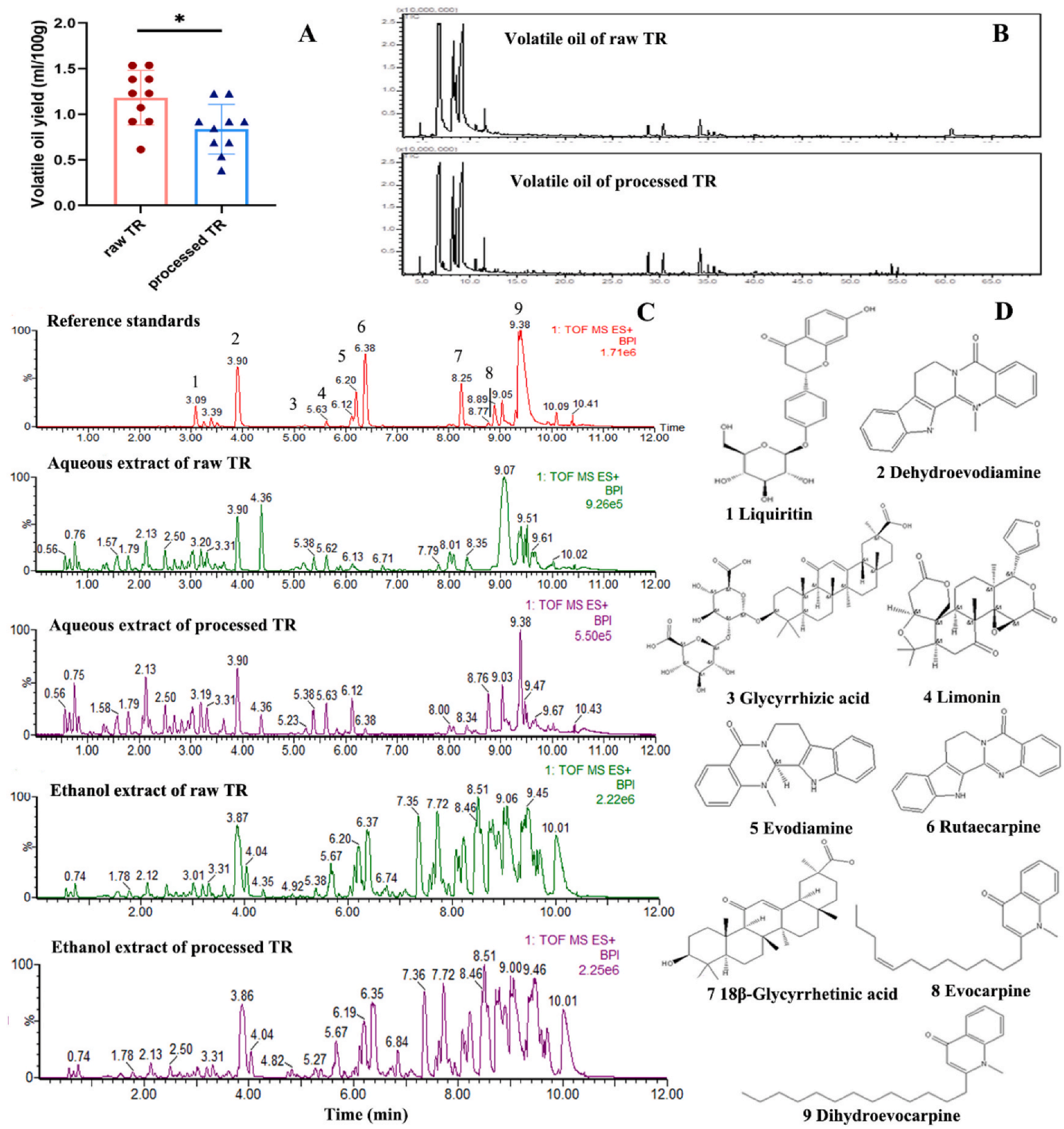

Fig. 3. The contents of the volatile oil of raw and processed *Tetradium ruticarpum* (TR, ml/100g, n=10), *$p < 0.05$, compared with the volatile oil yield of raw TR (A); total ion chromatography of volatile oil from raw and processed TR obtained by GC-MS (B); typical base peak intensity of reference standards, aqueous and ethanol extracts of TR acquired by UPLC-QTOF-MS in positive mode (C); chemical structures of reference standards (D).

Table 2
The fold change on raw aqueous extract *vs* raw ethanol extract of *Tetradium ruticarpum* (potential toxic component pool of *Tetradium ruticarpum*).

No.	RT (min)	Compound name	Formula	Molecular Weight (Da)	$[M+H]^+$ or $[M-H]^-$	Aqueous extract		Ethanol extract		Raw AE/EE fold change
						Raw	Processed	Raw	Processed	
1	4.36	6-acetoxy-5-epilimonin	$C_{28}H_{32}O_{10}$	528.1996	529.2074	424,564	61,770	109,138	4634	3.89
2	2.7	12α-hydroxylimonin	$C_{26}H_{30}O_9$	486.1890	487.1968	25,138	23,740	20,078	20,561	1.25
3	2.1	4-Methoxycoumarin	$C_{10}H_8O_3$	176.0473	175.0395	32,382	19,119	30,640	18,892	1.06
4	0.65	Quinic Acid	$C_7H_{12}O_6$	192.0634	191.0556	67,978	38,806	65,234	44,890	1.04

toxicokinetics, toxicodynamics, and the toxic mode of action.

Since the analysis of the extracts with LC/MS resulted in large quantities of identified compounds, and the toxicity tests of AE revealed to be more toxic than EE, the fold-change-filtering analysis was performed on raw AE/EE to effectively narrow down the pool of potential toxic ingredients. As shown in Table 2, four potential compounds were identified as 6-acetoxy-5-epilimonin, 12α-hydroxylimonin, and 4-methoxycoumarin and quinic acid. Among them, 6-acetoxy-5-epilimonin was pointed out to be the potential toxic ingredients in the previous study (Li et al., 2016). Its analogous, acetoxy-6-hydroxylimonin, was reported to dispose in hepatic tissue and may participate in lipid peroxidation, lipid transport, and fatty acid metabolism (https://hmdb.ca/metabolites/HMDB 0032834). Since 12α-hydroxylimonin was sharing the similar chemical structure with that component, therefore, it

Table 3
The fold change TOP 60 on processed ethanol extract *vs* processed aqueous extract of *Tetradium ruticarpum* (potential effective component pool).

No.	RT (min)	Compound name	Formula	Molecular Weight (Da)	$[M+H]^+$ or $[M-H]^-$	Aqueous extract		Ethanol extract		Pro EE/AE fold change
						Raw	Pro	Raw	Pro	
1	6.32	7β-hydroxyrutaecarpine	C18H13N3O2	303.1008	304.1086	nd	nd	1003547	507,895	∞
2	5.7	dihydrorutaecarpine iso	C18H15N3O	289.1215	290.1293	nd	nd	464,168	361,693	∞
3	7.57	1-methyl-2-[(1E,5Z)-1,5-undecadienyl]-4 (1H)-quinolone	C21H27NO	309.2093	310.2171	nd	nd	316,765	254,445	∞
4	7.68	1-methyl-2-[(1E,5Z)-1,5-undecadienyl]-4 (1H)-quinolone iso	C21H27NO	309.2093	310.2171	nd	nd	214,291	232,834	∞
5	7.85	1-methy-2-[(Z)-6-undecenyl]-4-(1H)-quinolone	C21H29NO	311.2249	312.2327	nd	nd	248,842	232,772	∞
6	8.17	evodianinine	C19H13N3O	299.1059	300.1137	nd	nd	141	208,567	∞
7	7.93	2-undecyl-4 (1H)-quinolinone	C20H29NO	299.2249	300.2327	nd	nd	373,341	196,291	∞
8	7.79	1-methyl-2-decyl-4(1H)-quinolone	C20H29NO	299.2249	300.2327	nd	nd	140,028	129,505	∞
9	8.1	1-methy-2-[(4Z,7Z)-4,7-tridecadienyl]-4(1H)-quinolone	C23H31NO	337.2406	338.2484	nd	nd	132,554	123,069	∞
10	6.71	wuchuyuamide I	C19H19N3O	305.1528	306.1606	nd	nd	124,202	114,914	∞
11	7.13	1-methyl-2-[(1E,5Z)-1,5-undecadienyl]-4(1H)-quinolone iso	C21H27NO	309.2093	310.2171	nd	nd	74,822	92,698	∞
12	5.99	rutaecarpine iso	C18H13N3O	287.1059	288.1137	nd	nd	nd	75,632	∞
13	4.75	9H-pyrido (3,4-b) indole, 3-(4-methoxybenzamidomethyl)-1-methyl-	C21H19N3O2	345.1477	346.1555	nd	nd	16,600	66,877	∞
14	6.19	14-formyldihydrorutaecarpine iso	C19H15N3O2	317.1164	318.1242	nd	nd	20,807	47,797	∞
15	6.33	1-methyl-2-[(Z)-7-tridecenyl]-4-(1H)-quinolone	C19H17N3O2	319.1321	320.1399	nd	nd	45,322	38,951	∞
16	5.44	7β-hydroxyrutaecarpine iso	C18H13N3O2	303.1008	304.1086	nd	nd	42,760	32,361	∞
17	6.19	1-methy-2-[(4Z,7Z)-4,7-tridecadienyl]-4(1H)-quinolone	C23H31NO	337.2406	338.2484	nd	nd	37,113	31,469	∞
18	5.57	7β-hydroxyrutaecarpine iso	C18H13N3O2	303.1008	304.1086	nd	nd	29,150	27,060	∞
19	8.25[a]	18β-glycyrrhetinic acid	C30H46O4	470.3396	471.3474	nd	nd	nd	18,598	∞
20	5.51	jangomolide	C26H28O8	468.1784	469.1862	nd	nd	615	6984	∞
21	6.98	dihydrorutaecarpine	C18H15N3O	289.1215	290.1293	nd	nd	165	4804	∞
22	5.67	14-formyldihydrorutaecarpine	C19H15N3O2	317.1164	318.1242	nd	3167	493,072	468,404	∞
23	5.29	dehydroevodiamine iso	C19H15N3O	301.1215	302.1293	nd	1372	5080	85,273	∞
24	8.23	1-methy-2-[(4Z,7Z)-4,7-tridecadienyl]-4(1H)-quinolone	C23H31NO	337.2406	338.2484	100	4026	1,724,541	1,674,439	17,196.50
25	7.36	1-methy-2-nonyl-4(1H)-quinolone	C19H27NO	285.2093	286.2171	113	3160	1,764,884	1,642,570	15,559.13
26	7.72	1-methyl-2-[(Z)-5-undecenyl]-4-(1H)-quinolone	C21H29NO	311.2249	312.2327	125	5480	1,856,100	1,785,190	14,827.52
27	6.2[a]	evodiamine	C19H17N3O	303.1372	304.1450	108	2593	1,225,574	1,177,511	11,334.97
28	8.2	1-methy-2-[(Z)-6-undecenyl]-4-(1H)-quinolone iso	C21H29NO	311.2249	312.2327	61	339	434,359	359,047	7161.12
29	8.51	1-methyl-2-undecyl-4(1H)-quinolone	C21H31NO	313.2406	314.2484	336	1979	2,219,618	2,165,675	6608.21
30	6.28	N-(4-ethoxyphenyl)-N,2-dimethyl-4-quinazolinamine	C18H19N3O	293.1528	294.1606	62	433	239,248	206,200	3869.57
31	9.7	1-methyl-2-tetradecyl-4(1H)-quinolone iso	C24H37NO	355.2875	356.2953	362	2576	1,194,114	1,070,080	3295.25
32	10.02	1-methy-2-pentadecyl-4-(1H) -quinolone	C25H39NO	369.3032	370.3110	1390	13,655	2,708,856	2,818,355	1949.03
33	8.43	1-methy-2-[(Z)-6-undecenyl]-4-(1H)-quinolone iso	C21H29NO	311.2249	312.2327	86	241	145,438	178,697	1685.54
34	6.38[a]	rutaecarpine	C18H13N3O	287.1059	288.1137	1187	20,282	1,788,282	1,849,108	1507.00
35	8.87	2-tridecyl-4-(1H) -quinolone	C22H33NO	327.2562	328.2640	643	1138	665,782	689,069	1035.86
36	9.01	1-methyl-2-dodecyl-4-(1H)-quinolone	C22H33NO	327.2562	328.2640	330	4049	332,031	326,122	1005.09
37	9.46	1-methyl-2-[(Z)-9-pentadecenyl]-4-(1H)-quinolone	C25H37NO	367.2875	368.2953	6384	93,513	3,556,793	3,480,047	557.14
38	8.9	1-methyl-2-[(Z)-7-tridecenyl]-4-(1H)-quinolone	C23H33NO	339.2562	340.2640	4909	12,121	2,079,410	2,027,539	423.62
39	9.17	1-methyl-2-tetradecenyl-4(1H)-quinolone	C24H35NO	353.2719	354.2797	1975	5227	470,417	416,483	238.15
40	9.58	1-methyl-2-tetradecyl-4(1H)-quinolone	C24H37NO	355.2875	356.2953	316	1198	58,607	31,464	185.51
41	9.03	1-methyl-2-[(6Z,9Z)-6,9-pentadecadienyl]-4(1H)-quinolone	C25H35NO	365.2719	366.2797	7151	188,356	1,286,416	1,230,494	179.89
42	6.85	7,8-dehydrorutaecarpine	C18H11N3O	285.0902	286.0980	120	529	13,943	367,164	116.24
43	9.53	dihydroevocarpine iso	C23H35NO	341.2719	342.2797	14,601	14,684	984,938	644,573	67.46
44	9.54	1-methyl-2-[(6Z,9Z)-6,9-pentadecadienyl]-4(1H)-quinolone	C25H35NO	365.2719	366.2797	1328	6591	83,698	80,740	63.03
45	4.62	rhetsinine (hydroxyevodiamine)	C19H17N3O2	319.1321	320.1399	191	712	10,197	7955	53.30
46	6.13	3-dimethyallyl-4-methoxy-2-quinolone	C15H17NO2	243.1259	244.1337	597	1049	28,329	25,946	47.46
47	9.38[a]	dihydroevocarpine	C23H35NO	341.2719	342.2797	123,393	393,275	3,269,291	3,140,323	26.49
48	6.37	obacunone	C26H30O7	454.1992	455.2070	884	855	22,234	18,595	25.14
49	5.14	12α-hydroxylimonin	C26H30O9	486.1890	485.1812	12,256	17,343	287,115	42,247	23.43
50	6.04	evodiamide	C19H21N3O	307.1685	308.1763	6404	9831	140,189	127,175	21.89
51	10.38	dihydroevocarpine iso	C23H35NO	341.2719	342.2797	1185	933	22,189	20,627	18.72
52	3.74	methyhydroxyevodiamine	C20H19N3O2	333.1477	334.1555	2222	1293	39,286	27,537	17.68
53	6.13	licoricone	C22H22O6	382.1416	383.1494	522	1259	7938	7666	15.21

(*continued on next page*)

Table 3 (*continued*)

No.	RT (min)	Compound name	Formula	Molecular Weight (Da)	$[M+H]^+$ or $[M-H]^-$	Aqueous extract		Ethanol extract		Pro EE/AE fold change
						Raw	Pro	Raw	Pro	
54	6.02	jangomolide	C26H28O8	468.1784	467.1706	13,338	13,228	131,593	158,432	9.87
55	6.14	6β-acetoxy-5-epilimonin/Glaucin B	C28H32O10	528.1996	527.1918	78,555	206,892	747,511	717,993	9.52
56	3.8	14-formyldihydrorutaecarpine iso	C19H15N3O2	317.1164	318.1242	2870	6045	27,283	101,366	9.50
57	6.13	12α-hydroxylimonin	C26H30O9	486.1890	485.1812	170,257	441,847	1,289,657	1,278,116	7.57
58	8.77[a]	evocarpine	C23H33NO	339.2562	340.2640	6179	145,306	42,931	39,858	6.95
59	4.93	skimmianine	C14H13NO4	259.0845	260.0923	6588	6956	43,376	34,296	6.58
60	3.87[a]	dehydroevodiamine	C19H15N3O	301.1215	302.1293	393,732	247,383	2,372,169	2,129,377	6.02

[a] Standard reference; nd: not detectable; na: not applicable.

Table 4
The fold change TOP 20 of *Tetradium ruticarpum* before and after processing.

No.	RT (min)	Compound name	Formula	Molecular Weight (Da)	$[M+H]^+$ or $[M-H]^-$	Aqueous extract		Fold change	Ethanol extract		Fold change
						Raw	Pro		Raw	Pro	
1	8.25[a]	18β-Glycyrrhetinic acid	C30H46O4	470.3396	471.3474	nd	nd	na	nd	18,598	∞
2	3.81	neoisoliquiritin	C21H22O9	418.1264	417.1186	nd	1440	∞	nd	6916	∞
3	5.54	glycyrrhizic acid iso	C42H62O16	822.4038	821.3960	nd	23,091	∞	nd	61,875	∞
4	5.23[a]	glycyrrhizic acid	C42H62O16	822.4038	821.3960	nd	184,305	∞	nd	378,260	∞
5	3.09[a]	liquiritin	C21H22O9	418.1264	417.1186	nd	18,596	∞	nd	25,973	∞
6	8.17	evodianinine	C19H13N3O	299.1059	300.1137	nd	nd	na	141	208,567	1482.09
7	5.26	evodiamine isomer	C19H17N3O	303.1372	304.1450	56	3383	60.92	219	114,364	523.21
8	4.97	glycyrrhizic acid derivatives	C42H62O17	838.3987	839.4065	nd	4133	∞	nd	7789	432.27
9	5.12	28-Glucosyl-30-methyl-3b,23-dihydroxy-12-oleanene-28,30-dioate 3-[arabinosyl-(1->3)-glucuronide]	C48H74O21	986.4723	987.4689	11	612	54.52	nd	1160	157.59
10	5.99	rutaecarpine iso	C18H13N3O	287.1059	288.1137	nd	nd	na	497	75,632	152.28
11	4.65	glycyrrhizic acid derivatives	C42H62O17	838.3987	839.4065	nd	1217	∞	nd	1918	118.49
12	9.92	N-[(4-heptoxy-3-methoxyphenyl)-[(3-methoxybenzoyl) amino] methyl]-3-methoxybenzamide	C31H38N2O6	534.2730	535.2711	184	226	1.23	2337	77,720	33.26
13	6.98	dihydrorutaecarpine	C18H15N3O	289.1215	290.1293	nd	nd	na	165	4804	29.10
14	6.85	7,8-dehydrorutaecarpine	C18H11N3O	285.0902	286.0980	120	529	4.41	13,943	367,164	26.33
15	4.36	6-acetoxy-5-epilimonin/Glaucin B	C28H32O10	528.1996	529.2074	424,564	61,770	6.87	109,138	4634	23.55
16	5.13	quercetin 3-(6″-caffeylgalactoside)	C30H26O15	626.1272	627.1382	245	335	1.37	4979	693	23.55
17	5.29	dehydroevodiamine iso	C19H15N3O	301.1215	302.1293	nd	1372	6.63	5080	85,273	16.79
18	3.14	12α-hydroxylimonin	C26H30O9	486.1890	487.1968	3999	718	5.57	6144	389	15.81
19	10.19	3,3'-((4R,5S,6S,7R)-tetrahydro-5,6-dihydroxy-2-oxo-4,7-bis (phenylmethyl)-1H-1,3-diazepine-1,3 (2H)-diyl) bis (methylene)bis (N-methyl-benzamide)	C42H40N2O3	620.3056	621.3134	45	39	1.15	225,567	16,659	13.54
20	5.51	jangomolide	C26H28O8	468.1784	469.1862	nd	nd	na	615	6984	11.35

[a] Standard reference; nd: not detectable; na: not applicable.

may attribute to steatosis in the liver in a same way. Besides, coumarins are already known chemicals with impacts on blood especially the coagulation system, which may explain the TR's toxic effect as ischemia of extremities. In addition, processing could decrease the contents of these four components in both AE and EE significantly (Table 2), which could lead to the attenuation in toxicity after processing in both AE and EE (Table 1). Therefore, the raw/processed fold change on AE extract could be valuable in identifying potential toxic components. It would be interesting to verify if these ingredients are the major source of TR's toxicity in future studies.

The potential therapeutic components were also screened by this fold-change-filtering strategy. Considering that the pharmacological effects of EE was more potent than AE (Fig. 2), the TOP 60 of Pro EE vs Pro AE in Table 3 gave us a profile of bioactive compounds of TR on pain-relieving and inflammation. The processing procedure demonstrated a significant improvement of its therapeutic effects in mice in the EE group, whereas showed less impact on mice in the AE group. Interestingly, most of the identified components were indole and quinolone-like alkaloids, along with scattered limonoids, amides and 18β-glycyrrhetinic acid. The bioactivity of those ingredients were proved in the previous pharmacological studies (Matsuda et al., 1997, 1998; Nie et al., 2016; Wang et al., 2018). Compared with aqueous extraction, extraction by 70% ethanol could dramatically increase the content of most bioactive constituents in TR, particularly as alkaloids, the major bioactive ingredient of TR (Tang et al., 1996b), and 18β-glycyrrhetinic acid, the proved anti-inflammatory chemical agent (Kim et al., 2018; Seckl et al., 1993; Su, L. et al., 2018a). Dehydroevodiamine, one major bioactive component from TR, was substantially increased when extracted by 70% ethanol. The previous study indicated that dehydroevodiamine could inhibit LPS-induced iNOS and COX-2 gene expression in RAW 264.7 macrophages by blocking the NF-κB activation (Noh et al., 2006), proved as a strong anti-inflammatory compound. The increased solubility of these bioactive components in EE (compared with AE) may be the major contributors to its stronger analgesic and anti-inflammation activities.

After the processing procedure of stir-frying with licorice extract, TR demonstrated better performance of therapeutic effects on all three classical pharmacological models. The following fold-change analysis gave us a comprehensive profiling of the impacts on efficacy caused by exogenously imported chemicals (the licorice components) and some types of alkaloids with improved solubility after processing (Table 4). As a major ingredient of licorice, 18β-glycyrrhetinic acid is a corticosteroid-like analog with an extraordinary anti-inflammatory effect through inhibiting adrenal sulfotransferase activity or regulating the mitochondrial ROS/MAPK axis (Al-Dujaili et al., 2011; Kim et al., 2018; Seckl et al., 1993; Su, L. et al., 2018a), which may partly explain the improved efficacy after processing. 6 acetoxy-5-epilimonin and hdyroxylimonin were decreased sharply after processing (Tables 2 and 4), which may due to deacetylation or dehydroxylation reaction during heating. Therefore, these compounds may be the main contributors of its enhanced anti-inflammatory effect and attenuated toxicity in the processed TR product. Following separation and validation work of these ingredients with delineating the underlying mechanism of toxicity and efficacy would be an interesting next step to expand our knowledge of this valuable herbal remedy.

Interestingly, mice in the AE group demonstrated less potency in analgesic and anti-inflammatory effects whereas relatively higher risk of toxicity before processing when comparing with mice in either VO or EE group. Considering that the AE of TR is the most frequently used "aqueous decoction" form in clinics, the efficacy are to be questioned, whereas the 70% ethanol extraction and the standard processing procedure could provide a better rational and safety solution of TR's application.

To sum up, acute toxicity, efficacy, and component variations of TR have been studied in VO, AE, and EE before and after processing. TR demonstrated alleviated pungent smell, lower toxicity and improved efficacy after processing. The underlying processing mechanism was elucidated from a component perspective. The sharply decreased volatile oil content after processing explained the relieved acrid smell of raw TR. Potential toxic and bioactive constituents were also pointed out within this study, to better understand the reduced toxicity and enhanced efficacy after processing. It would be interesting to separate and purify them, followed by further exploration to see which gene or target is related with TR's toxicity or efficacy. Since processed TR is the major application form in clinics, our result would be valuable for rational drug use, safety guarantee and proofs of the value and importance of the TR processing procedure.

5. Conclusions

In summary, the integrated toxicity and efficacy analyses of volatile, aqueous and ethanol extracts of TR indicated that the processing procedure could effectively reduce its acute toxicity in all three extracts and enhance its analgesic and anti-inflammatory effects in volatile and ethanol extracts. The promising candidate compounds related to the toxicity and efficacy of TR were also identified. The results could expand our understanding of the value of the standard processing procedure of TR, be valuable to the quality control of TR manufacturing and administration, as well as support clinical rational and safety applications of this medicinal plant.

Author's contribution

Luping Qin and Gang Cao conceived the experiments; Guanqun Chen supervised on manuscript structure building and writing; Qiyuan Shan, Gang Tian, Hui Hui, Min Hao, Kuilong Wang and Xin Wu carried out the experiments; Qiyang Shou and Huiying Fu gave advices on animal studies; Qiyuan Shan, Gang Tian and Juli Wang wrote the manuscript. All the authors approved the final version of the manuscript prior to submission.

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Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2020.113292.

References

Al-Akhras, M.A.H., Aljarrah, K., Al-Khateeb, H., Jaradat, A., Al-Omari, A., Al-Nasser, A., Masadeh, M.M., Amin, A., Hamza, A.A., Mohammed, K., Al Olama, M., Daoud, S., 2012. Introducing Cichorium Pumilum as a potential therapeutical agent against drug-induced benign breast tumor in rats. Electromagn. Biol. Med. 31 (4), 299–309. https://doi.org/10.3109/15368378.2012.662193.

Al-Dabbagh, B., Elhaty, I.A., Al Hrout, A., Al Sakkaf, R., El-Awady, R., Ashraf, S.S., Amin, A., 2018. Antioxidant and anticancer activities of Trigonella foenum-graecum, Cassia acutifolia and Rhazya stricta. BMC Compl. Alternative Med. 18 (1), 240. https://doi.org/10.1186/s12906-018-2285-7.

Al-Dabbagh, B., Elhaty, I.A., Elhaw, M., Murali, C., Al Mansoori, A., Awad, B., Amin, A., 2019. Antioxidant and anticancer activities of chamomile (Matricaria recutita L.). BMC Res. Notes 12 (1), 3. https://doi.org/10.1186/s13104-018-3960-y.

Al-Dujaili, E.A.S., Kenyon, C.J., Nicol, M.R., Mason, J.I., 2011. Liquorice and glycyrrhetinic acid increase DHEA and deoxycorticosterone levels in vivo and in vitro by inhibiting adrenal SULT2A1 activity. Mol. Cell. Endocrinol. 336 (1–2), 102–109. https://doi.org/10.1016/j.mce.2010.12.011.

Al-Hrout, A., Baig, B., Hilal-Alnaqbi, A., Amin, A., 2017. Cancer and biotechnology: a matchup that should never slowdown. In: Biotechnology and Production of Anti-cancer Compounds. Springer, Cham.

Al-Hrout, A., Chaiboonchoe, A., Khraiwesh, B., Murali, C., Baig, B., El-Awady, R., Tarazi, H., Alzahmi, A., Nelson, D.R., Greish, Y.E., Ramadan, W., Salehi-Ashtiani, K., Amin, A., 2018. Safranal induces DNA double-strand breakage and ER-stress-mediated cell death in hepatocellular carcinoma cells. Sci. Rep. 8 (1), 16951. https://doi.org/10.1038/s41598-018-34855-0.

Amin, A., 2008. Ketoconazole-induced testicular damage in rats reduced by Gentiana extract. Exp. Toxicol. Pathol. 59 (6), 377–384. https://doi.org/10.1016/j.etp.2007.10.008.

Ashktorab, H., Soleimani, A., Singh, G., Amr, A., Tabtabaei, S., Latella, G., Stein, U., Akhondzadeh, S., Solanki, N., Gondré-Lewis, M.C., Habtezion, A., Brim, H., 2019. Saffron: the golden spice with therapeutic properties on digestive diseases. Nutrients 11 (5). https://doi.org/10.3390/nu11050943.

Cai, B.C., 2008. Processing on Chinese Materia Medica. Chinese Medicine Press, Beijing.

Cai, Q., Wei, J., Zhao, W., Shi, S., Zhang, Y., Wei, R., Zhang, Y., Li, W., Wang, Q., 2014. Toxicity of Evodiae fructus on rat liver mitochondria: the role of oxidative stress and mitochondrial permeability transition. Molecules 19 (12), 21168–21182. https://doi.org/10.3390/molecules191221168.

Cai, X., Meng, N., Yang, B., 2006. Analysis of a case of poisoning caused by overdose of Evodia rutaecarpa. Beijing J. Trad. Chin. Med. 25 (3), 171–172.

Chinese Pharmacopoeia Commission, 2015. Pharmacopeia of the People's Republic of China (2015 Edition) Part I, 2015. Chinese medical science and technology press.

Editorial committee of Flora of China, 2008. Flora of China. http://foc.iplant.cn/content.aspx?TaxonId=242351597. (Accessed 21 May 2019).

Ekor, M., 2014. The growing use of herbal medicines: issues relating to adverse reactions and challenges in monitoring safety. Front. Pharmacol. 4, 177. https://doi.org/10.3389/fphar.2013.00177.

El-Dakhly, S.M., Salama, A.A.A., Hassanin, S.O.M., Yassen, N.N., Hamza, A.A., Amin, A., 2020. Aescin and diosmin each alone or in low dose- combination ameliorate liver damage induced by carbon tetrachloride in rats. BMC Res. Notes 13 (1), 259. https://doi.org/10.1186/s13104-020-05094-2.

Finney, D.J., 1985. The median lethal dose and its estimation. Arch. Toxicol. 56 (4), 215–218.

Gong, P.Y., Tian, Y.S., Guo, Y.J., Gu, L.F., Li, J.Y., Qi, J., Yu, B.Y., 2019. Comparisons of antithrombosis, hematopoietic effects and chemical profiles of dried and rice wine-processed Rehmanniae Radix extracts. J. Ethnopharmacol. 231, 394–402. https://doi.org/10.1016/j.jep.2018.10.025.

Hamza, A.A., Ahmed, M.M., Elwey, H.M., Amin, A., 2016. Melissa officinalis protects against doxorubicin-induced cardiotoxicity in rats and potentiates its anticancer

activity on MCF-7 cells. PLoS ONE 11 (11), e0167049. https://doi.org/10.1371/journal.pone.0167049.
Hamza, A.A., Heeba, G.H., Elwy, H.M., Murali, C., El-Awady, R., Amin, A., 2018. Molecular characterization of the grape seeds extract's effect against chemically induced liver cancer: in vivo and in vitro analyses. Sci. Rep. 8 (1), 1270. https://doi.org/10.1038/s41598-018-19492-x.
Hamza, A.A., Mohamed, M.G., Lashin, F.M., Amin, A., 2020. Dandelion prevents liver fibrosis, inflammatory response, and oxidative stress in rats. J. Basis Appl. Zoo. 81 (1), 43. https://doi.org/10.1186/s41936-020-00177-9.
Hayes, A.W., Loomis, T.A., 1996. Loomis's Essentials of Toxicology. Elsevier.
Hosseinzadeh, H., Ramezani, M., Salmani, G.-a., 2000. Antinociceptive, anti-inflammatory and acute toxicity effects of Zataria multiflora Boiss extracts in mice and rats. J. Ethnopharmacol. 73 (3), 379–385. https://doi.org/10.1016/s0378-8741(00)00238-5.
Huan, T., Palermo, A., Ivanisevic, J., Rinehart, D., Edler, D., Phommavongsay, T., Benton, H.P., Guijas, C., Domingo-Almenara, X., Warth, B., Siuzdak, G., 2018. Autonomous multimodal metabolomics data integration for comprehensive pathway analysis and systems biology. Anal. Chem. 90 (14), 8396–8403. https://doi.org/10.1021/acs.analchem.8b00875.
Huang, W., Sun, R., 2012. Study on analgesic effect and safety scope of different components of Evodia Fructus. Chin. J. Pharmaco. 9 (7), 397–400. https://doi.org/10.3969/j.issn.1672-8629.2012.07.004.
Huang, W., Zhao, Y., Sun, R., 2010. Experimental comparison study on mice's acute toxicity of different composition in Evodia fructus. Chin. J. Pharmaco. 7 (3), 129–134. https://doi.org/10.3969/j.issn.1672-8629.2010.03.001.
Hui, H., Zhou, H.Q., Xu, X.F., Zhou, R.J., Shan, Q.Y., Qin, L.P., 2019. Optimization of the extraction process of the volatile oil in Evodia rutaecarpa. Chin. Archive TCM. http://kns.cnki.net/kcms/detail/21.1546.r.20191210.20191702.20191134.html.
Iwata, H., Tezuka, Y., Kadota, S., Hiratsuka, A., Watabe, T., 2005. Mechanism-based inactivation of human liver microsomal CYP3A4 by rutaecarpine and limonin from Evodia fruit extract. Drug Metabol. Pharmacokinet. 20 (1), 34–45.
Kamal, H., Jafar, S., Mudgil, P., Murali, C., Amin, A., Maqsood, S., 2018. Inhibitory properties of camel whey protein hydrolysates toward liver cancer cells, dipeptidyl peptidase-IV, and inflammation. J. Dairy Sci. 101 (10), 8711–8720. https://doi.org/10.3168/jds.2018-14586.
Kim, Y.H., Kim, D.E., Lee, S.-H., 2018. Effects of 18-glycyrrhetinic acid on fungal protease-induced airway inflammatory responses. Mediat. Inflamm. 2018, 6461032. https://doi.org/10.1155/2018/6461032.
Ko, H.C., Wang, Y.H., Liou, K.T., Chen, C.M., Chen, C.H., Wang, W.Y., Chang, S., Hou, Y. C., Chen, K.T., Chen, C.F., 2007. Anti-inflammatory effects and mechanisms of the ethanol extract of Evodia rutaecarpa and its bioactive components on neutrophils and microglial cells. Eur. J. Pharmacol. 555 (2), 211–217. https://doi.org/10.1016/j.ejphar.2006.10.002.
Li, W., Sun, X., Liu, B., Zhang, L., Fan, Z., Ji, Y., 2016. Screening and identification of hepatotoxic component in Evodia rutaecarpa based on spectrum-effect relationship and UPLC-Q-TOFMS. Biomed. Chromatogr. 30 (12), 1975–1983. https://doi.org/10.1002/bmc.3774.
Li, X.Y., Wu, X.W., Dou, L.W., Yin, L.S., Huang, W., Sun, R., 2015a. Research of oxidative damage mechanism on hepatotoxicity caused by volatile oil from evodia fructus in mice. Chin. J. Pharmaco. 12 (1), 16–19.
Li, Z.Y., Sun, H.M., Xing, J., Qin, X.M., Du, G.H., 2015b. Chemical and biological comparison of raw and vinegar-baked Radix Bupleuri. J. Ethnopharmacol. 165, 20–28. https://doi.org/10.1016/j.jep.2015.02.024.
Mahady, G.B., 2001. Global harmonization of herbal health claims. J. Nutr. 131 (3), 1120S–1123S. https://doi.org/10.1093/jn/131.3.1120S.
Matsuda, H., Wu, J.X., Tanaka, T., Iinuma, M., Kubo, M., 1997. Antinociceptive activities of 70% methanol extract of evodiae fructus (fruit of Evodia rutaecarpa var. bodinieri) and its alkaloidal components. Biol. Pharm. Bull. 20 (3), 243–248.
Matsuda, H., Yoshikawa, M., Iinuma, M., Kubo, M., 1998. Antinociceptive and anti-inflammatory activities of limonin isolated from the fruits of Evodia rutaecarpa var. bodinieri. Planta Med. 64 (4), 339–342.
Nair, A.B., Jacob, S., 2016. A simple practice guide for dose conversion between animals and human. J. Basic Clin. Pharm. 7 (2), 27–31. https://doi.org/10.4103/0976-0105.177703.
Newman, D.J., Cragg, G.M., 2020. Natural products as sources of new drugs over the nearly four decades from 01/1981 to 09/2019. J. Nat. Prod. 83 (3), 770–803. https://doi.org/10.1021/acs.jnatprod.9b01285.
National Research Council, 2011. Guide for the Care and Use of Laboratory Animals, eighth ed. National Academy Press, Washington, DC.
Nie, X.Q., Chen, H.H., Zhang, J.Y., Zhang, Y.J., Yang, J.W., Pan, H.J., Song, W.X., Murad, F., He, Y.Q., Bian, K., 2016. Rutaecarpine ameliorates hyperlipidemia and hyperglycemia in fat-fed, streptozotocin-treated rats via regulating the IRS-1/PI3K/Akt and AMPK/ACC2 signaling pathways. Acta Pharmacol. Sin. 37 (4), 483–496. https://doi.org/10.1038/aps.2015.167.
Noh, E.J., Ahn, K.S., Shin, E.M., Jung, S.H., Kim, Y.S., 2006. Inhibition of lipopolysaccharide-induced iNOS and COX-2 expression by dehydroevodiamine through suppression of NF-kappaB activation in RAW 264.7 macrophages. Life Sci. 79 (7), 695–701. https://doi.org/10.1016/j.lfs.2006.02.020.
Pommier, Y., 2006. Topoisomerase I inhibitors: camptothecins and beyond. Nat. Rev. Canc. 6 (10), 789–802. https://doi.org/10.1038/nrc1977.
Salem, G.A., Alamyel, F.B., Abushaala, F.A., Hussain, M.S., Elnory, k.A., Abusheba, H., Sahu, R.P., 2019. Evaluation of the hepatoprotective, anti-inflammatory, antinociceptive and antiepileptic activities of Chrysanthemum trifurcatum. Biomed. Pharmacother. 117, 109123. https://doi.org/10.1016/j.biopha.2019.109123.
Seckl, J.R., Dow, R.C., Low, S.C., Edwards, C.R., Fink, G., 1993. The 11 beta-hydroxysteroid dehydrogenase inhibitor glycyrrhetinic acid affects corticosteroid feedback regulation of hypothalamic corticotrophin-releasing peptides in rats. J. Endocrinol. 136 (3), 471–477.
Shan, B., Cai, Y.Z., Brooks, J.D., Corke, H., 2007. Antibacterial properties and major bioactive components of cinnamon stick (Cinnamomum burmannii): activity against foodborne pathogenic bacteria. J. Agric. Food Chem. 55 (14), 5484–5490.
Shan, Q.Y., Cao, G., Cai, H., Cong, X.D., Cai, B.C., 2012. Novel software-based method to classify structurally similar compounds combined with high performance liquid chromatography–quadrupole time of flight mass spectrometry to identify complex components of herbal medicines. J. Chromatogr. A 1264, 13–21. https://doi.org/10.1016/j.chroma.2012.09.045.
Shan, Q.Y., Sang, X.N., Hui, H., Shou, Q.Y., Fu, H.Y., Hao, M., Liu, K.H., Zhang, Q.Y., Cao, G., Qin, L.P., 2020. Processing and polyherbal formulation of Tetradium ruticarpum (A. Juss.) hartley: phytochemistry, pharmacokinetics, and toxicity. Front. Pharmacol. 11 (133), 1–8. https://doi.org/10.3389/fphar.2020.00133.
Shoji, N., Umeyana, A., Iuchi, A., Saito, N., Arihara, S., Nomoto, K., Ohizumi, Y., 1989. Two novel alkaloids from evodia rutaecarpa. J. Nat. Prod. 52 (5), 1160–1162. https://doi.org/10.1021/np50065a043.
Shoji, N., Umeyana, A., Iuchi, A., Saito, N., Arihara, S., Takemoto, T., 1988. Isolation of an amide, a possible key precursor to evodiamine, from Evodia rutaecarpa. J. Nat. Prod 51 (1), 161–162. https://doi.org/10.1021/np50055a028.
Solanki, H.K., Shah, D.A., Maheriya, P.M., Patel, C.A., 2015. Evaluation of anti-inflammatory activity of probiotic on carrageenan-induced paw edema in Wistar rats. Int. J. Biol. Macromol. 72, 1277–1282.
Su, L., Wang, Z., Huang, F., Lan, R.L., Chen, X.Y., Han, D.P., Zhang, L.R., Zhang, W.J., Hong, J.S., 2018a. 18β-Glycyrrhetinic acid mitigates radiation-induced skin damage via NADPH oxidase/ROS/p38MAPK and NF-κB pathways. Environ. Toxicol. Pharmacol. 60, 82–90. https://doi.org/10.1016/j.etap.2018.04.012.
Su, X.L., Xu, S., Shan, Y., Yin, M., Chen, Y., Feng, X., Wang, Q.Z., 2018b. Three new quinazolines from Evodia rutaecarpa and their biological activity. Fitoterapia 127, 186–192. https://doi.org/10.1016/j.fitote.2018.02.003.
Sugimoto, T., Miyase, T., Kuroyanagi, M., Ueno, A., 1988. Limonoids and quinolone alkaloids from evodia rutaecarpa BENTHAM. Chem. Pharm. Bull. 36 (11), 4453–4461.
Tang, Y.Q., Feng, X.Z., Huang, L., 1996a. Quinolone alkaloids from evodia rutaecarpa. Phytochemistry 43 (3), 719–722.
Tang, Y.Q., Feng, X.Z., Huang, L., 1996b. Quinolone alkaloids from evodia rutaecarpa. Acta Pharmacol. Sin. 34 (2), 151–155. https://doi.org/10.3321/j.issn:1001-5302.2004.02.017.
Tao, Y., Jiang, E., Cai, B., 2020. A biochemometrics strategy combining quantitative determination, bioactivity evaluation and relationship analysis for identification of analgesic alkaloids of raw and vinegar-processed Corydalis turtschaninovii. J. Separ. Sci. 43 (6), 1183–1189. https://doi.org/10.1002/jssc.201901021.
Tu, Y.Y., 2011. The discovery of artemisinin (qinghaosu) and gifts from Chinese medicine. Nat. Med. 17 (10), 1217–1220. https://doi.org/10.1038/nm.2471.
Upadhyay, P., Shukla, R., Mishra, S.K., 2019. Acute and sub-acute toxicity study of hydro-alcoholic leaves extract of Reinwardtia indica in rats. Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie 111, 36–41. https://doi.org/10.1016/j.biopha.2018.12.056.
Wang, L., Dou, L.W., Guo, W., Li, X.Y., Yang, Q., Wu, X.W., Sun, R., 2017. Toxic Q-marker identification based on conventional application of traditional Chinese medicine: taking Evodiae Fructus as an example. Chin. Tradit. Herb. Drugs 48 (6), 1159–1166. https://doi.org/10.7501/j.issn.0253-2670.2017.06.017.
Wang, S.C., Yang, Y., Liu, J., Jiang, A.D., Chu, Z.X., Chen, S.Y., Gong, G.Q., He, G.W., Xu, Y.G., Zhu, Q.H., 2018. Discovery of novel limonin derivatives as potent anti-inflammatory and analgesic agents. Chin. J. Nat. Med. 16 (3), 231–240. https://doi.org/10.1016/S1875-5364(18)30052-9.
Wang, Z.R., Cui, Y.Y., Vainstein, A., Chen, S.W., Ma, H.Q., 2017. Regulation of fig (Ficus carica L.) fruit color: metabolomic and transcriptomic analyses of the flavonoid biosynthetic pathway. Front. Plant Sci. 8 (1990) https://doi.org/10.3389/fpls.2017.01990.
Wilhelm, S., 2006. Treatment of Cancer with Sorafenib. Bayer Healthcare LLC.
Xu, Y., Cai, H., Cao, G., Duan, Y., Pei, K., Tu, S., Zhou, J., Xie, L., Sun, D., Zhao, J., Liu, J., Wang, X., Shen, L., 2018. Profiling and analysis of multiple constituents in Baizhu Shaoyao San before and after processing by stir-frying using UHPLC/Q-TOF-MS/MS coupled with multivariate statistical analysis. J. Chromatogr. B 1083, 110–123. https://doi.org/10.1016/j.jchromb.2018.03.003.
Yang, X., Wang, H., Ni, H.-M., Xiong, A., Wang, Z., Sesaki, H., Ding, W.-X., Yang, L., 2017. Inhibition of Drp 1 protects against senecionine-induced mitochondria-mediated apoptosis in primary hepatocytes and in mice. Redox Biol. 12, 264–273. https://doi.org/10.1016/j.redox.2017.02.020.
Zhang, X.F., Gao, N.N., Liu, Y.H., Du, H., Yang, L., Li, F., 2011. A study on the composition and toxicity changes of essential oil of evodia rutaecarpa (Juss.) benth. Before and after processing. Pharm. J. Chin. PLA. 27 (3), 229–232. https://doi.org/10.3969/j.issn.1008-9926.2011.03.014.
Zhou, L., Xu, J.D., Zhou, S.S., Shen, H., Mao, Q., Kong, M., Zou, Y.T., Xu, Y.Y., Xu, J., Li, S.L., 2017. Chemomics-based marker compounds mining and mimetic processing for exploring chemical mechanisms in traditional processing of herbal medicines, a continuous study on Rehmanniae Radix. J. Chromatogr. A 1530, 232–240. https://doi.org/10.1016/j.chroma.2017.11.036.